

WARRUWI, Australia

WMO: 941390

Lat: 11.6503S

Lon: 133.3797E

Elev: 65

StdP: 14.66

Time Zone: 9.50 (AUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	68.1	69.7	47.9	49.4	74.8	51.2	56.0	76.3	20.3	78.1	18.9	77.5	9.0	130	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	9.7	92.6	78.5	91.7	78.5	90.8	78.4	83.4	86.2	81.8	86.8	80.9	86.9	8.8	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.9	172.1	84.1	80.6	159.1	84.2	79.2	151.7	84.5	47.6	85.9	45.6	86.9	44.6	87.1	91.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	16.7	15.6	DB	65.1	95.2	2.1	1.0	63.6	95.9	62.4	96.4	61.2	97.0	59.7	97.7
			WB	54.7	83.2	1.8	1.1	53.5	84.0	52.4	84.6	51.4	85.2	50.2	86.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	83.9	83.5	83.3	83.1	81.4	78.0	77.2	77.8	80.8	83.7	85.2	85.5	(6)
	DBStd	3.49	2.24	2.23	1.97	1.82	2.04	2.80	2.23	1.99	1.95	1.75	1.41	2.41	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11653	1051	937	1032	992	973	840	842	862	923	1044	1057	1100	(10)
	CDD65	6178	586	517	567	542	508	390	377	397	473	579	607	635	(11)
	CDH74	67492	7178	6100	6613	6330	5275	2996	2517	2857	4572	6852	7807	8395	(12)
(13)	CDH80	24374	2900	2275	2415	2332	1530	578	368	490	1230	2665	3573	4018	(13)
(14)	Wind	WSAvg	8.0	7.7	7.4	6.3	8.1	10.3	10.2	9.3	8.0	7.2	5.9	5.7	(14)
(15) Precipitation	PrecAvg	46.9	11.1	10.1	9.5	5.0	0.8	0.1	0.1	0.0	0.1	0.4	2.7	7.7	(15)
	PrecMax	70.5	23.0	19.8	21.5	25.7	7.3	1.8	1.7	0.2	0.6	2.6	18.0	16.7	(16)
	PrecMin	24.0	2.3	1.3	1.5	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.5	(17)
	PrecStd	12.3	4.6	4.8	5.5	5.3	1.3	0.3	0.3	0.0	0.1	0.6	3.5	4.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.7	91.9	92.2	91.5	89.7	87.8	86.4	87.0	89.6	92.3	93.3	94.5	(19)
		MCWB	80.1	79.8	80.2	78.0	76.4	74.2	71.1	70.1	73.6	74.9	77.4	78.5	(20)
	2%	DB	91.3	90.4	90.2	89.9	88.1	86.0	84.6	85.3	88.0	90.7	91.9	92.9	(21)
		MCWB	79.8	79.9	79.7	77.6	75.9	73.5	70.8	70.0	73.4	76.0	77.6	78.9	(22)
	5%	DB	90.0	89.3	89.0	88.8	87.1	84.6	83.4	84.0	86.7	89.6	91.0	91.8	(23)
		MCWB	79.8	79.6	79.5	77.3	75.5	72.9	70.7	70.0	73.4	76.2	77.5	78.8	(24)
	10%	DB	88.6	88.2	87.9	87.8	86.0	83.3	82.2	82.8	85.5	88.6	90.0	90.7	(25)
		MCWB	79.7	79.4	79.4	77.0	75.1	72.2	70.4	69.8	73.3	76.1	77.3	78.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	84.2	82.1	87.5	80.7	80.0	80.5	79.9	75.8	78.2	84.7	80.7	83.3	(27)
		MCDB	87.2	89.1	88.1	87.6	84.9	80.6	80.2	82.1	85.4	84.8	89.0	86.6	(28)
	2%	WB	82.4	81.1	84.2	79.8	78.7	76.4	75.6	74.6	77.0	79.1	79.9	81.7	(29)
		MCDB	86.9	87.7	84.7	86.4	85.1	82.0	79.0	80.7	83.7	86.5	88.1	87.5	(30)
	5%	WB	81.2	80.6	81.5	79.2	77.9	75.5	74.1	73.8	76.1	78.2	79.2	80.8	(31)
		MCDB	86.9	87.1	85.9	85.4	84.2	81.7	79.3	79.9	83.0	86.2	87.4	87.8	(32)
	10%	WB	80.4	80.0	80.4	78.7	77.2	74.6	73.0	73.0	75.3	77.5	78.6	80.1	(33)
		MCDB	86.7	86.1	85.9	84.8	83.5	81.1	79.1	79.3	82.5	85.5	86.8	87.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	8.7	9.2	9.7	9.9	11.0	11.0	11.3	10.8	10.4	10.7	9.7	(35)
		MCDBR	10.6	10.1	10.6	10.6	10.4	11.0	11.4	12.0	11.6	11.5	11.4	10.9	(36)
	5% WB	MCWBR	4.0	4.0	4.3	3.8	4.5	5.7	6.1	6.1	4.5	4.0	3.3	3.6	(37)
		MCWBR	9.5	9.3	9.5	9.4	9.2	9.9	10.1	10.0	10.2	10.2	10.6	10.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.415	0.407	0.396	0.379	0.372	0.361	0.354	0.365	0.409	0.427	0.426	0.413	(40)
		taud	2.520	2.538	2.572	2.608	2.598	2.583	2.594	2.541	2.411	2.379	2.425	2.507	(41)
	Ebn at Noon		294	295	293	288	280	279	283	288	285	286	289	294	(42)
		Edh at Noon	36	35	33	30	29	29	29	32	38	41	39	36	(43)
(44) All-Sky Solar Radiation	RadAvg		1656	1719	1681	1733	1662	1600	1682	1905	2099	2187	2123	1816	(44)
	RadStd		202	233	195	200	110	56	53	53	70	85	82	190	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.72	N/A	N/A	N/A	N/A	-0.85	N/A	N/A	N/A	N/A

Nomenclature: See separate page